From Ghost Echoes to Useful Geometry: Covariance-Aware Virtual-Aperture Radar Tracking

Extended research manuscript

Abstract

Specular radar echoes can provide additional observation directions, but their benefit depends on coordinate consistency, uncertain reflectors and correct measurement reuse. We derive the equivalence of sensor-image and target-image rays, explain a 15.189-degree receiving-frame mismatch, and propagate reflector distance and orientation uncertainty into coupled range/range-rate covariance. A fixed-prior iterated update relinearizes without repeatedly consuming the same evidence. In an 80-seed independent-truth range/rate tracking study, mean trajectory RMSE decreases from 2.858 m for direct observations to 0.304 m with a virtual leg and approximately 0.284 m with reflector uncertainty propagation. Diagonal and coupled propagation are nearly indistinguishable in this fixture. An earlier complete-chain prototype and 155 instrumented real-data replay records are reported separately, with metric and modeling limitations made explicit.

1 Introduction

A guardrail echo is inconvenient precisely because it is structured. It often follows the same target through several scans, carries a stable path geometry and changes Doppler with the target's motion. Deleting it removes a false object from a conventional tracker, but can also discard a second direction from which the real object has been observed. The challenge is to recover that information without turning every coincidental echo into evidence. We organize the problem around a simple chain: estimate the reflecting geometry, test candidate paths against kinematics, and decide how much information each admitted observation should contribute. The last step matters as much as the first. A duplicated or overconfident reflection can make the filter less accurate even when its geometric interpretation looks plausible. This paper develops a receiving-frame-consistent image model, its observability and reflector-covariance consequences, and a fixed-prior iterated update that consumes each observation once. A matched 80-seed simulation uses independent truth geometry and compares direct-only tracking with three virtual-aperture covariance policies. We also reproduce a historical estimate–gate–assimilate experiment, but identify its metric and model limitations rather than promote its 67% figure into a fleet-accuracy claim. Instrumented replay counters from 155 real-data records provide a separate integration check.

2 Related Work

Multipath exploitation is not a new premise. Virtual-anchor positioning and tracking already use reflected paths as information, with uncertainty modeling central to reliable inference [6]. In automotive MIMO radar, Zheng *et al.* exploit the difference between departure and arrival angles to detect multipath ghosts using a GLRT and continuous-domain angular estimation [5]. That signal-level detection problem is distinct from updating a target state with an already hypothesized reflected observation. Kalman filtering [1] supplies the

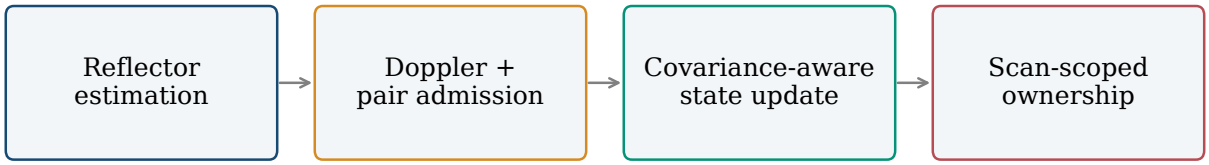


Figure 1: The proposed processing chain separates path discovery, physical admission, statistical weighting and observation ownership. An echo is not useful merely because it resembles a reflection.

estimation machinery, while IMM tracking [2] addresses maneuver-model uncertainty. Adaptive CFAR [4] addresses clutter-dependent detection. Our contribution is not to replace these foundations or claim that image-source geometry was previously incorrect. It is to make the coordinate map, reflector uncertainty and measurement lifecycle explicit in an automotive tracking formulation and evaluate their consequences under controlled comparisons.

3 Geometry: Two Correct Images, One Receiving Frame

Let a planar reflector satisfy $\mathbf{n}^T\mathbf{x} = d$ with unit normal $\mathbf{n}$. Define $H = I - 2\mathbf{n}\mathbf{n}^T$ and

$$F(\mathbf{x}) = H\mathbf{x} + 2d\mathbf{n}. \tag{1}$$

For sensor $\mathbf{s}$ and target $\mathbf{p}$, their images are $\mathbf{s}^* = F(\mathbf{s})$ and $\mathbf{p}^* = F(\mathbf{p})$. The two unfolded rays obey

$$\mathbf{p}^* - \mathbf{s} = H(\mathbf{p} - \mathbf{s}^*). \tag{2}$$

Because $H^T H = I$, their lengths agree exactly. Their coordinates are not interchangeable as physical arrival bearings: the Householder map must be applied when returning an unfolded ray to the receiving frame.

3.1 The 15.19-degree example, stated precisely

In Fig. 2, the unfolded sensor-image direction is $(30, -4)$ and the target-image direction is $(30, 4)$. Confusing their coordinates gives

$$\Delta\theta = 2\arctan(4/30) = 15.1893^\circ. \tag{3}$$

At the longitudinal plane $x = 30$ m, the corresponding ray endpoints differ by 8 m laterally. This is a frame-mismatch example, not a universal defect in target mirroring and not a constant error at arbitrary 30 m slant range. The propagation class must also be specified. A reflected one-way leg has length $\rho_s = \|\mathbf{p} - \mathbf{s}^*\|$. A mixed direct/reflected radar round trip has total length $\rho_d + \rho_s$ and conventional apparent range $(\rho_d + \rho_s)/2$. A symmetric two-reflected-leg path has apparent range ρ_s. The numerical study below evaluates a calibrated virtual-leg observation model; applying it to an unclassified mixed path would introduce model error.

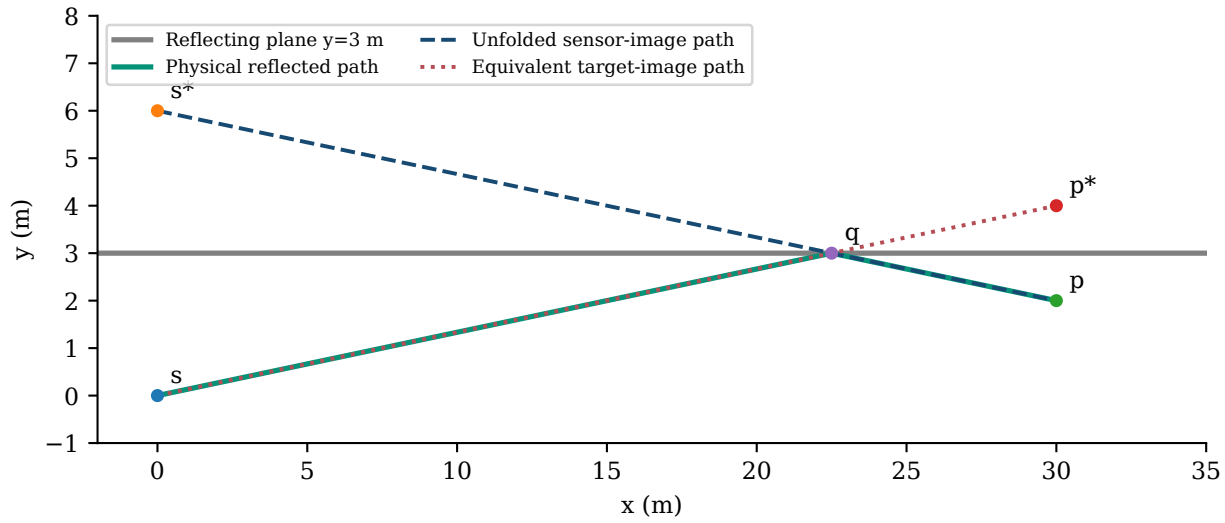


Figure 2: Planar example with sensor $(0,0)$ m, wall $y = 3$ m and target $(30,2)$ m. The physical contact is $(22.5,3)$ m. Sensor-image and target-image constructions have the same length and are related by a reflection; using one ray's coordinates as the other's bearing creates a sign error.

3.2 What the displacement invariant does and does not prove

For an origin-centred sensor, define $\mathbf{q}_s = \mathbf{p} - 2d\mathbf{n}$. Then

$$\mathbf{p} - \mathbf{q}_s = 2d\mathbf{n}. \tag{4}$$

Several independent estimates of $\mathbf{q}_s$ for a common plane share a rank-one displacement direction. If $\mathbf{q}_s$ is constructed from the same assumed $(d,\mathbf{n})$, however, the equality is an identity rather than an association test. A useful residual must compare independent observations after a physically justified coordinate conversion. Ordinary target-image differences instead have magnitude $2|\mathbf{n}^T\mathbf{p} - d|$, which varies with target location.

4 A Virtual Aperture and Its Information

For state $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$, origin-centred sensor and static reflector, define

$$\rho_d = \|\mathbf{p}\|, \qquad \dot{\rho}_d = \mathbf{u}_d^T\mathbf{v}, \tag{5}$$

$$\rho_s = \|\mathbf{p} - 2d\mathbf{n}\|, \qquad \dot{\rho}_s = \mathbf{u}_s^T\mathbf{v}, \tag{6}$$

where $\mathbf{u}_d = \mathbf{p}/\rho_d$ and $\mathbf{u}_s = (\mathbf{p} - 2d\mathbf{n})/\rho_s$. A parent and its reflection generally do not share a Doppler bin: $\Delta\dot{\rho} = \mathbf{v}^T(\mathbf{u}_s - \mathbf{u}_d)$. The project fixture produces a 1.987 m/s difference for a lateral target velocity of 10 m/s; that is large compared with its 0.05 m/s radial-speed noise.

4.1 When does the second direction help?

The range-position Jacobian is $U = [\mathbf{u}_d^T; \mathbf{u}_s^T]$. In two dimensions,

$$\det(U^TU) = \sin^2\psi, \tag{7}$$

where ψ is the angle between the directions. With equal independent range noise σ^2, the normalized information eigenvalues are $1 \pm |\cos\psi|$. Two nearly parallel paths add mostly radial information; they do not

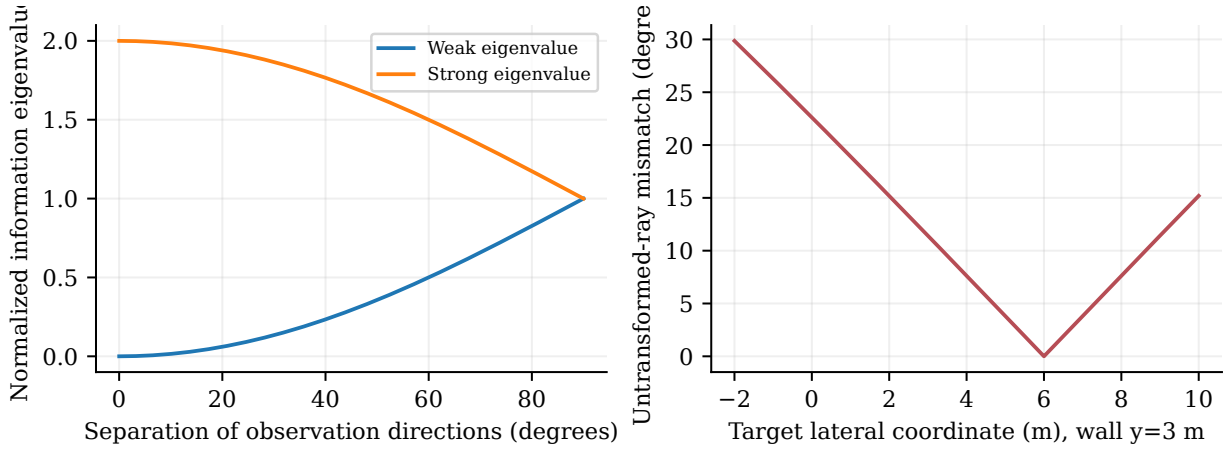


Figure 3: Analytical geometry diagnostics. Left: information eigenvalues expose the weak direction as the two paths become parallel. Right: untransformed bearing mismatch varies with target position; it is not a universal constant.

magically supply strong lateral accuracy. The second direction becomes useful when its angular separation is appreciable and its noise model remains credible.

5 Reflector Uncertainty and a Single-Use Update

Parameterize a planar normal as $\mathbf{n} = (\cos\phi, \sin\phi)^T$ with tangent $\mathbf{t} = (-\sin\phi, \cos\phi)^T$. For $\xi = (d, \phi)^T$ and independent reflector uncertainty Σ_ξ, linearization gives

$$R_{\text{eff}} = R_0 + J_\xi \Sigma_\xi J_\xi^T. \tag{8}$$

The specular-range derivatives are $\partial\rho_s/\partial d = -2\mathbf{u}_s^T\mathbf{n}$ and $\partial\rho_s/\partial\phi = -2d\mathbf{u}_s^T\mathbf{t}$. Let $\mathbf{w} = (\mathbf{v} - \dot{\rho}_s\mathbf{u}_s)/\rho_s$; the corresponding rate derivatives are $-2\mathbf{w}^T\mathbf{n}$ and $-2d\mathbf{w}^T\mathbf{t}$. Range and range rate therefore share reflector-induced covariance. Inflating their diagonal entries independently discards that correlation.

5.1 Iterate the linearization, not the evidence

Let $(\mathbf{x}^-, P^-)$ be the predicted prior. At iteration j, compute $h_j = h(\mathbf{x}_j)$, H_j and R_j, then

$$K_j = P^- H_j^T (H_j P^- H_j^T + R_j)^{-1}, \tag{9}$$

$$\mathbf{x}_{j+1} = \mathbf{x}^- + K_j[\mathbf{z} - h_j + H_j(\mathbf{x}_j - \mathbf{x}^-)]. \tag{10}$$

The prior covariance remains fixed during these iterations. After the final linearization, update covariance once using the Joseph form. Repeatedly applying an ordinary measurement update and shrinking covariance inside the same iteration loop treats the same data as fresh independent evidence. The distinction is consequential when range-rate linearization couples a poor velocity prior into a position innovation.

5.2 Ownership and suppression

Every admitted observation receives an identifier scoped to sensor, scan and detection. Assimilation marks that identifier consumed. Downstream multipath suppression must not penalize the same evidence after the target update used it. The implementation's negative sentinel is one representation of this contract; an explicit

enum is another. The research requirement is the lifecycle, not the numerical value of a private field. A reused detection must be excluded or modeled with its correct correlation.

6 From a Plane to a Curved Barrier

Let $\mathbf{q} = \mathbf{s} + t\mathbf{u}$ be a specular contact, $\delta = \mathbf{p} - \mathbf{s}$, and $\ell = t + \|\delta - t\mathbf{u}\|$ the broken-path length. Rearrangement yields

$$t = \frac{\ell^2 - \|\delta\|^2}{2(\ell - \mathbf{u}^T \delta)}. \tag{11}$$

Require a unit $\mathbf{u}$, positive denominator and $0 < t < \ell$. Grazing and collinear configurations are poorly conditioned even when the formula returns a finite number. Two contacts with tangent slopes determine a local cubic Hermite graph. Its confluent Vandermonde determinant is $(x_2 - x_1)^4$, establishing uniqueness for distinct abscissae but not numerical stability. A shifted, scaled coordinate is preferable. This graph is a local approximation, not an exact clothoid.

7 Association Under Maneuvers

For a candidate residual $\mathbf{y}$ with innovation covariance S, the whitened statistic is $D^2 = \mathbf{y}^T S^{-1} \mathbf{y}$. A bounded kinematic policy may use

$$\gamma = \mathrm{clip}_{[\gamma_0, \gamma_{\max}]}\{\gamma_0(1 + c_a a_n + c_e D^2/m)\}, \tag{12}$$

where a_n is normalized acceleration and m is measurement dimension. Unlike a ratio of already-whitened D^2 to a dimensional covariance trace, each added term is dimensionless. The policy depends on the tested candidate, so a fixed chi-square coverage statement no longer applies automatically. For $c = \gamma_0 c_e/m < 1$, its unsaturated acceptance inequality becomes $D^2 \le \gamma_0(1 + c_a a_n)/(1 - c)$; if $c \ge 1$, it admits every candidate until the outer clamp acts. This identifies an "open-door" failure analytically. An IMM can predict mode probabilities before association [2]. A narrow common gate may nevertheless starve a maneuver mode of evidence. That is a useful onset stress test, not a universal refutation of IMM tracking. The archived adaptive-gate prototype also propagates the tracker once per attempted candidate, which confounds candidate order with elapsed time; its 99.6% result is consequently retained as exploratory rather than used as the headline comparison here.

8 Experimental Evaluation

8.1 Independent-truth covariance study

We simulate a two-dimensional constant-velocity target from $\mathbf{x}_0 = (25, 1, 4, 0.4)^T$, a true wall at $d = 3$ m and $\phi = \pi/2$, 60 scans at 20 Hz, range noise 0.10 m and radial-speed noise 0.05 m/s. The filter starts with component perturbation standard deviations $(1, 1, 2, 2)$ and matching diagonal covariance. Discrete process noise is $GG^T(0.3)^2$, with $G = [\frac{1}{2}\Delta t^2 I; \Delta t I]$. Each scan supplies an external reflector estimate with distance standard deviation 0.15 m and angle standard deviation 2°. Measurements are generated from the true reflector, never from the estimate supplied to the filter. Four methods share each seed's initialization and measurement draws: direct only, dual with sensor-noise covariance only, dual with diagonal reflector propagation, and dual with full coupled propagation. Every filter uses three fixed-prior iterations and one covariance update. For each seed we calculate $\sqrt{K^{-1}\sum_k \|\hat{\mathbf{p}}_k - \mathbf{p}_k\|^2}$, then average this RMSE across 80 seeds. Confidence intervals bootstrap complete seeds, not correlated individual scans. The mean RMSE is 2.858 m for direct-only tracking, 0.304 m for dual observations with sensor noise only, 0.284 m for diagonal

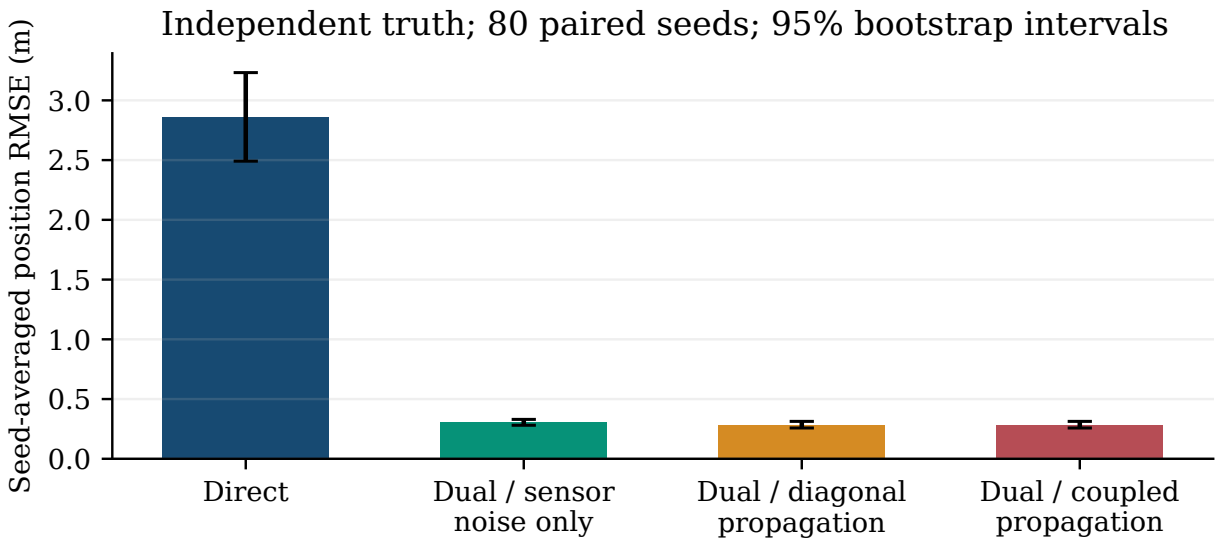


Figure 4: Matched-seed independent-truth comparison. Error bars are 95% percentile bootstrap intervals over complete seeds. The diagonal and coupled policies are nearly indistinguishable in this fixture; the data do not support a large full-covariance gain over diagonal propagation.

Table 1: Independent-truth position RMSE across 80 seeds

Method	Mean (m)	95% interval (m)
Direct only	2.858	[2.491, 3.232]
Dual / sensor noise	0.304	[0.281, 0.330]
Dual / diagonal	0.284	[0.258, 0.313]
Dual / coupled	0.284	[0.258, 0.313]

uncertainty propagation and 0.284 m for coupled propagation. The result supports the value of the additional path in this deliberately weak direct-only range/rate geometry. It does not imply the same gain over a tracker already using accurate angle measurements. Nor does it establish superiority over a modern automotive tracking baseline.

8.2 Reproducing the earlier complete-chain result

The existing 40-seed experiment recovers edge recall/precision 0.969/0.902 and mean Euclidean position error 1.969 to 0.651 m. The distance/normal errors average 0.260 m and 3.282°. Three details limit interpretation. First, the reported position statistic is mean Euclidean error, not RMSE. Second, the tracking leg generates ghost observations using the supplied estimated reflector, so it does not expose the complete geometric mismatch to the filter. Third, it repeats covariance updates inside a three-step loop. We report this result to connect the prototype history to the present controlled study, not as independent validation of a robust end-to-end fleet system.

8.3 Instrumented real-data replay

The 155 paired replay records contain 71,779 detections, 9,478 OFF versus 9,131 ON association events, and 10,130 ON assimilation marks in 151 records. These are reproducible counters, not labeled association

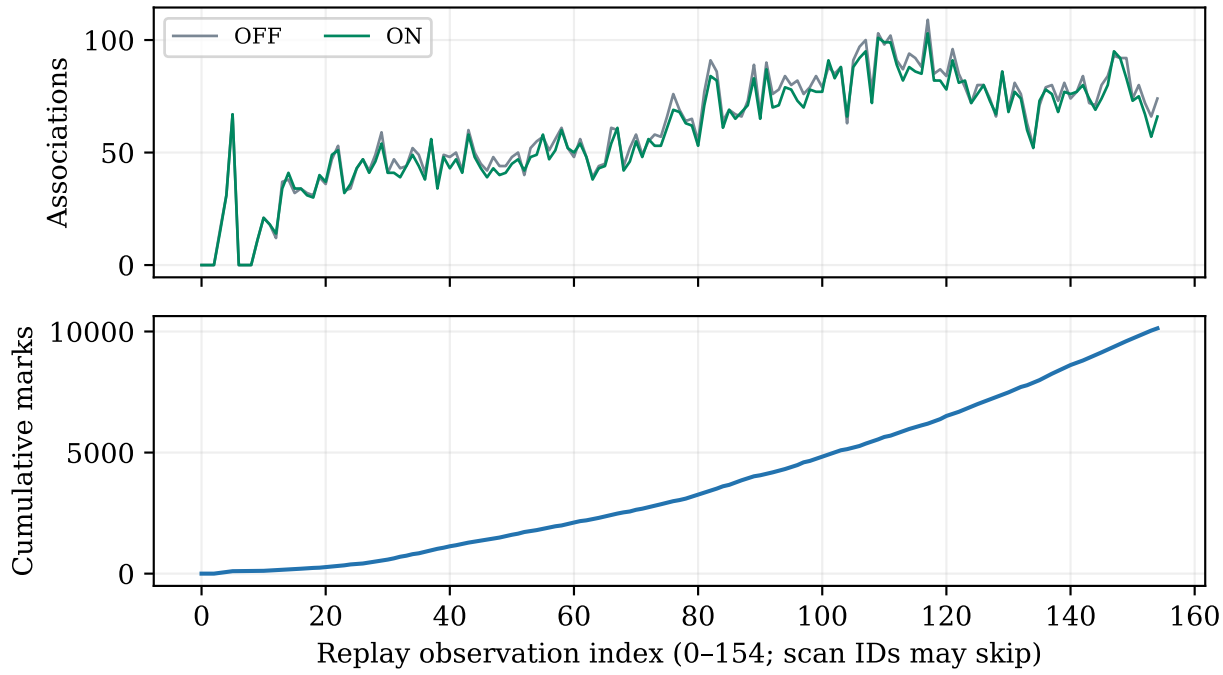


Figure 5: Actual instrumented replay: association counts and cumulative assimilation marks. Scan IDs are not consecutive; the horizontal coordinate is the paired record index. This figure tests event accounting, not ground-truth tracking accuracy.

accuracy. The harness constructs pairing candidates from geometric proximity rather than independently annotated reflection paths. Neither position truth nor recall denominators are present in those CSVs.

9 Discussion and Conclusion

Multipath is valuable when it supplies an additional, sufficiently different observation direction and when uncertainty follows the geometry into the update. A correct image construction alone is not enough. Receiving-frame conversion, independent truth generation, single-use evidence and a clear distinction between detection counters and target accuracy are equally necessary. The strongest result here is a transparent controlled study: virtual-leg observations reduce error in a range/rate-only tracking geometry, while reflector uncertainty modestly improves their weighting. Full covariance is mathematically justified, but its incremental benefit over diagonal propagation is small in the tested fixture. Curved surfaces, mixed propagation classes, uncertain association and comparison with an angle-enabled production tracker remain the meaningful next tests. That is a sharper foundation for a conference paper than attributing every improvement in a large software stack to a single impressive percentage.

A Propagation Classes and Coordinate Conventions

A.1 A path model before a filter model

A radar's apparent range is determined by round-trip delay, not by whichever Euclidean segment is most convenient in an image diagram. If $L(\mathbf{p})$ is the total propagation length, apparent monostatic range is $L/2$

and apparent radial speed is $\dot{L}/2$. The direct path has $L = 2\rho_d$. A mixed path with one direct leg and one reflected leg has $L = \rho_d + \rho_s$. A symmetric two-reflected-leg construction has $L = 2\rho_s$. Their Jacobians and Doppler predictions differ even when the same wall image appears in the geometry. For the mixed path,

$$\nabla_{\mathbf{p}} r_{\text{mix}} = \tfrac{1}{2}(\mathbf{u}_d + \mathbf{u}_s), \quad \dot{r}_{\text{mix}} = \tfrac{1}{2}(\mathbf{u}_d + \mathbf{u}_s)^T \mathbf{v}. \tag{13}$$

The controlled experiment uses separate calibrated leg observations, with the direct and virtual-leg noises stated explicitly. Applying its result directly to every unclassified multipath detection would conceal this propagation distinction. A signal-level path classifier or an explicit mixture model belongs upstream of the proposed update.

A.2 Proof of image equivalence

Proposition 1 (Receiving-frame image identity)**.** *For $F(\mathbf{x}) = H\mathbf{x} + 2d\mathbf{n}$, $H = I - 2\mathbf{n}\mathbf{n}^T$ and unit $\mathbf{n}$, $F(\mathbf{p}) - \mathbf{s} = H[\mathbf{p} - F(\mathbf{s})]$.*

Proof. The right side is $H\mathbf{p} - H^2\mathbf{s} - 2dH\mathbf{n}$. Since $H^2 = I$ and $H\mathbf{n} = -\mathbf{n}$, it equals $H\mathbf{p} - \mathbf{s} + 2d\mathbf{n} = F(\mathbf{p}) - \mathbf{s}$. □

Norm equality follows from orthogonality. The identity does not say that the two ray-coordinate vectors have the same angle; it says which transformation makes them equivalent. This is the proper explanation of a range-exact, bearing-mismatched implementation.

A.3 The local rank-one construction

When a virtual-leg vector is independently reconstructed, displacement consensus can identify a common reflector direction. The translation $2d\mathbf{n}$ has two unknown scalar degrees of freedom in a planar scene. A magnitude-only search may appear to reduce dimensionality, but estimating direction from all pairs inside a magnitude window can admit a large set of accidental pairings. The historical comparison illustrates this: a one-dimensional magnitude strategy had recall/precision 0.845/0.727, whereas its own direction-search baseline reached 0.993/0.909. The negative result argues for joint geometric testing, not for discarding the displacement relation.

B Observability and Aperture Conditioning

B.1 Information eigenvalues

Let $\mathbf{u}_1$ and $\mathbf{u}_2$ be unit directions with inner product c. The nonzero eigenvalues of $U^T U$ are the eigenvalues of $UU^T = \left[\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right]$, namely $1 \pm |c|$. Thus a second nearly parallel path does not repair the weak transverse direction. For small separation ψ, $1 - \cos\psi \simeq \psi^2/2$: the weak information grows only quadratically with aperture angle. With unequal independent observation variances, information becomes $U^T R^{-1} U$. A noisy reflected path may supply less useful transverse information than a cleaner direct bearing observation. This is why the main result is scoped to a range/rate-only baseline. An angle-enabled baseline is the important next comparator for a radar conference evaluation.

B.2 Velocity information and position–velocity coupling

For $\rho = \|\mathbf{q}\|$, $\mathbf{u} = \mathbf{q}/\rho$ and $v_r = \mathbf{u}^T \mathbf{v}$, differentiation gives

$$\frac{\partial v_r}{\partial \mathbf{p}} = \frac{\mathbf{v} - v_r \mathbf{u}}{\rho}, \qquad \frac{\partial v_r}{\partial \mathbf{v}} = \mathbf{u}^T. \tag{14}$$

The first derivative is large when transverse speed is high and range is short. Ignoring it makes a position error look like a radial-speed anomaly. An iterated linearization can help, but only if it does not reuse the observation as fresh evidence on every iteration.

B.3 Information gain does not guarantee lower realized error

Under an exact linear Gaussian model and independent observations, adding a measurement cannot increase posterior covariance. Realized estimation error under model mismatch can still increase. An uncertain wall, incorrect association, wrong path class or duplicated observation invalidates the assumptions behind that monotonic covariance argument. A smaller covariance is therefore not itself proof of a better tracker; accuracy and consistency should be scored separately.

C Reflector Jacobians in Planar and Spatial Models

C.1 Planar distance-angle representation

Write $\mathbf{q} = \mathbf{p} - 2d\mathbf{n}(\phi)$. Then $\partial\mathbf{q}/\partial d = -2\mathbf{n}$ and $\partial\mathbf{q}/\partial\phi = -2d\mathbf{t}$. For a scalar function $f(\mathbf{q})$, $\partial f/\partial\xi = (\nabla_{\mathbf{q}} f)^T[-2\mathbf{n}, -2d\mathbf{t}]$. Substituting range and range-rate gradients yields the derivatives used in the main study. The representation handles the unit-normal constraint intrinsically. Treating both normal components as unconstrained independent parameters would introduce an unphysical radial direction in normal space. In three dimensions, a local two-dimensional tangent basis B with $B^T\mathbf{n} = 0$ gives $\delta\mathbf{n} = B\delta\zeta$ and covariance $B\Sigma_\zeta B^T$. Isotropic tangent uncertainty is $\sigma_n^2(I - \mathbf{n}\mathbf{n}^T)$.

C.2 State–reflector correlation

Equation (8) assumes the reflector perturbation is independent of the predicted target error. If the same target detections helped estimate the reflector, that assumption can fail. Let $C = \mathrm{Cov}(\delta\mathbf{x}, \delta\xi)$. The innovation covariance contains additional terms

$$S = H_x P H_x^T + J_\xi \Sigma_\xi J_\xi^T + H_x C J_\xi^T + J_\xi C^T H_x^T + R_0. \tag{15}$$

Marginal inflation without these cross terms is an approximation. A joint target/reflector state, a Schmidt-style nuisance treatment or a carefully separated external reflector estimate is needed when that coupling is important. Our controlled study uses an external, independently perturbed reflector estimate precisely to make the simpler model's assumptions testable.

C.3 Across-scan dependence

A static wall estimate persists across scans. If its error is fixed but the filter treats it as newly independent noise at every scan, confidence can accumulate too rapidly. The present fixture samples an independent external estimate per scan; it does not validate that static-nuisance case. A follow-up experiment should hold a reflector error fixed across a trajectory and compare marginal-noise inflation with a joint-state estimator. That is a distinct experiment, not an unreported benefit of the current one.

D Fixed-Prior Iteration Derived from a Local MAP Objective

For a predicted Gaussian state and a nonlinear observation, the local objective is

$$\mathscr{L}(\mathbf{x}) = \tfrac{1}{2}\left\|\mathbf{x} - \mathbf{x}^-\right\|^2_{(P^-)^{-1}} + \tfrac{1}{2}\left\|\mathbf{z} - h(\mathbf{x})\right\|^2_{R^{-1}}. \tag{16}$$

Linearize h at $\mathbf{x}_j$ while retaining the same prior. Minimizing the resulting quadratic gives Eq. (10). The correction term $H_j(\mathbf{x}_j - \mathbf{x}^-)$ is what keeps the changing linearization anchored to the original prior rather than treating $\mathbf{x}_j$ as a new Bayesian prior.

D.1 Covariance update

After the final iteration, with final gain K and Jacobian H, use

$$P^+ = (I - KH)P^-(I - KH)^T + KRK^T. \tag{17}$$

This Joseph expression preserves positive semidefiniteness in exact arithmetic and is generally more robust to numerical roundoff than subtractive forms. Explicit symmetrization removes small asymmetric roundoff but does not turn an invalid repeated update into a Joseph update.

D.2 State-dependent covariance

The reflector Jacobian makes R_{eff} depend on the linearization point. Recomputing it during the three fixed-prior iterations is a practical local approximation. A full likelihood with state-dependent covariance also contains a log-determinant term and derivatives of R; the present update does not optimize that complete objective. Stating this approximation is more informative than claiming generic optimality from the word "iterated."

E Duplicate Evidence and the Assimilation Contract

E.1 A two-observation example

Two scalar observations with equal variance and correlation c have covariance $R = \sigma^2 \left[\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right]$. Their information about a shared scalar mean is

$$\mathbf{1}^T R^{-1} \mathbf{1} = \frac{2}{\sigma^2(1+c)}. \tag{18}$$

As $c \to 1$, this tends to the information of a single observation, not twice that amount. Appending the same range-rate sample as a second diagonal row creates false information. A shared-reflector covariance term does not automatically model that duplicate sensor noise; observation identity is still necessary.

E.2 A scan-scoped state machine

The lifecycle has four conceptual states: unseen, candidate, admitted and consumed. Suppression may operate on an unconsumed candidate, while an admitted measurement that actually updates the target becomes consumed. A failed update should not leave a stale consumed flag. At scan close-out, temporary ownership is resolved before the public log is serialized. The exact representation is an implementation detail. A negative sentinel in a legacy probability field is compact but requires every relevant consumer to understand the contract and every log conversion to handle it. An explicit type is clearer in a new interface. Neither choice proves correct execution without testing exception paths, resets, sensor interleaving and detection-index reuse.

E.3 Two-scan confirmation needs two epochs

A counter keyed only by detection index may count two calls in one scan as two confirmations. It may also transfer history to a different object when that index is reused. A real temporal rule needs an identity and

an epoch, with at most one transition per pair per scan. These are algorithmic requirements, not merely software-style preferences, because they define what statistical evidence the confirmation rule represents.

F Contact Inversion: Derivation and Sensitivity

Starting from $\ell - t = \|\delta - t\mathbf{u}\|$, square both sides and cancel t^2 using $\|\mathbf{u}\| = 1$. This produces the rational expression in Eq. (11). Define $B = \ell - \mathbf{u}^T\delta$ and $\rho = \ell - t$. Its useful differential identities include

$$\frac{\partial t}{\partial \ell} = \frac{\rho}{B}, \qquad \frac{\partial t}{\partial \mathbf{p}} = \frac{t\mathbf{u} - \delta}{B}. \tag{19}$$

For a tangent perturbation $d\mathbf{u}$ preserving unit norm to first order,

$$dt = \frac{\rho}{B}d\ell + \frac{(t\mathbf{u} - \delta)^T}{B}d\mathbf{p} + \frac{t\delta^T}{B}d\mathbf{u}. \tag{20}$$

The $1/B$ factor is an explicit conditioning warning. A finite nominal contact does not justify an arbitrary noise tolerance when B is small.

F.1 Recovering the local normal

Let $\mathbf{a} = (\mathbf{p} - \mathbf{q})/\rho$ point from the contact to the target. Reflection geometry gives a normal parallel to $\mathbf{a} - \mathbf{u}$. Thus

$$\hat{\mathbf{n}} = \frac{\mathbf{a} - \mathbf{u}}{\|\mathbf{a} - \mathbf{u}\|}. \tag{21}$$

When $\mathbf{a} \approx \mathbf{u}$, the normal is ill-conditioned. The normalization Jacobian contains $(I - \mathbf{n}\mathbf{n}^T)/\|\mathbf{a} - \mathbf{u}\|$, so grazing configurations amplify noise independently of the path-length denominator.

F.2 Moving sensor and target

For a stationary reflector and a valid specular contact, differentiating the broken-path length eliminates the contact's tangential velocity through Fermat stationarity. The first derivative is

$$\dot{\ell} = \mathbf{a}^T\mathbf{v}_p - \mathbf{u}^T\mathbf{v}_s. \tag{22}$$

This is useful because explicit surface curvature does not enter the first derivative. It does not eliminate uncertainty in the contact directions, nor does it mean that a curved wall may be replaced by a globally fixed image plane.

G Hermite Barrier Geometry Without a Clothoid Claim

Given two contacts (x_1, y_1), (x_2, y_2) and slopes m_1, m_2, use $h = x_2 - x_1$ and $t = (x - x_1)/h$. The cubic is

$$y(t) = (2t^3 - 3t^2 + 1)y_1 + (-2t^3 + 3t^2)y_2 \tag{23}$$

$$+ (t^3 - 2t^2 + t)hm_1 + (t^3 - t^2)hm_2. \tag{24}$$

This normalized basis avoids large monomial powers of a world-coordinate abscissa. The determinant of the unscaled confluent Vandermonde matrix is h^4; its nonzero value establishes uniqueness, not immunity to noisy nearly coincident contacts.

G.1 Curvature and validity radius

For a graph $y(x)$, curvature is $\kappa = y''/(1+y'^2)^{3/2}$. A clothoid instead specifies curvature as an affine function of arc length. Calling every cubic graph a clothoid confuses a local interpolation model with a physically specific curve family. A local cubic can still be useful if its fitted interval, residuals and curvature bounds are reported. The historical planar method lost recall on a bent-wall scene, and a Hermite contact prototype showed non-monotone gate acceptance across some noise rungs. Those outcomes motivate covariance-weighted fitting and explicit validity intervals. They do not establish that a two-contact cubic has solved arbitrary highway infrastructure geometry.

H Bounded Gating and Contamination

H.1 Dimensionless statistics

The squared Mahalanobis distance D^2 is dimensionless. Dividing it by $\operatorname{tr}(S)$ adds inverse measurement units and makes a threshold depend on the arbitrary scale of the measurement representation. A dimensionless alternative uses D^2/m, while the simpler Euclidean ratio $\|\mathbf{y}\|^2/\operatorname{tr}(S)$ is meaningful only when residual components have compatible units. Mixing metres and m/s in that Euclidean ratio is not a harmless notational shortcut.

H.2 The candidate-dependent feedback inequality

Before clipping, Eq. (12) has the form $\gamma = A + cD^2$. Acceptance requires $(1-c)D^2 \le A$. If $c < 1$, this is a finite expanded gate. If $c \ge 1$ and $A > 0$, every nonnegative D^2 satisfies the unsaturated condition. The outer ceiling then does all the rejection work. This algebra explains why a seemingly modest innovation coefficient can turn an adaptive rule into a fixed maximum gate. An empirical recall increase in that regime does not by itself demonstrate intelligent kinematic adaptation. A fair comparison must include the same fixed maximum gate, clutter precision, track-error statistics and a single prediction per scan. The archived prototype does not satisfy all those requirements, so it is not used as definitive evidence for the revised formula.

H.3 Frozen rank calibration

For clean sorted scores $S_{(1)} \le \cdots \le S_{(n)}$, replacing at most b samples can move an order statistic by at most b ranks, subject to valid boundary indices. A contaminated selected rank k is sandwiched between clean ranks $k-b$ and $k+b$. The prototype's $(231, 247)$ interval corresponds to an eight-replacement budget around rank 239 for a 255-score set. That is a deterministic rank statement, not automatic online coverage under distribution drift. Freezing calibration prevents a sliding window from absorbing a burst of outliers as normal. It also prevents legitimate adaptation to a new environment. A useful evaluation must report both contamination robustness and performance after real distribution changes, rather than treating freezing as unconditionally superior.

I Environmental Front-End Support

I.1 Spectral uncertainty scaling

For normalized nonnegative spectral powers p_i, entropy is $H = -\sum p_i \log p_i$. The proposed covariance scale $\exp(\beta H/\mathrm{SNR})$ reduces measurement influence when entropy is high and linear SNR is low. It needs a finite lower SNR bound and an upper scale or an explicit overflow policy. The exponential has no finite

ceiling by itself. An identity-default entropy feed is a useful integration choice but does not validate wet-weather accuracy. The historical spray fixture reports mean Euclidean error 0.911 to 0.315 m in a controlled burst model; translating that into fleet rain performance requires labeled recordings and consistent sensor measurements.

I.2 Weighted CFAR and exact false-alarm assumptions

For independent exponential reference cells and prior-fixed nonnegative weights β_i, a weighted threshold has false-alarm expression $P_{\mathrm{FA}} = \prod_i (1+\alpha\beta_i)^{-1}$. If the weights are learned from the same current cells, that expression no longer follows from the same conditional independence argument. Tracker-predicted occupancy can provide prior information, but its feedback correlation with the detector should be made explicit. The archived occupancy-BLUE analysis derives $\beta_i \propto m_i/v_i$ with normalization $\sum_i \beta_i m_i = 1$. This is an unbiased minimum-variance weighting under its moment model, not a universal robust clutter estimator. The production overlay's all-free occupancy feed implements only a narrower special case.

J Additional Experimental Interpretation

J.1 What the paired bootstrap means

The uncertainty intervals in the main text resample complete seeds 2,000 times. They quantify variation of the mean across this fixed synthetic scene family. They do not measure uncertainty about unseen wall types, path classes or road conditions. Reporting those intervals as a fleet confidence bound would exceed their scope. The diagonal and coupled policies share almost identical means and strongly overlapping intervals. The correct conclusion is that this fixture does not resolve a substantial incremental benefit from the cross-covariance. That is compatible with deriving the full covariance as the appropriate model: an effect can be mathematically necessary yet small in a particular operating geometry.

J.2 Mean error, RMSE and aggregation

For nonnegative errors e_k, mean Euclidean error is $K^{-1}\sum_k e_k$ and RMSE is $\sqrt{K^{-1}\sum_k e_k^2}$. Averaging per-seed RMSE differs again from pooling all squared errors before the square root. A table must identify its aggregation rather than call every metre-valued statistic "RMSE." The revised comparison uses per-seed trajectory RMSE followed by a seed average; the earlier chain values remain explicitly mean Euclidean error.

J.3 Real-data counters as a systems result

The paired replay counters are useful because they show how many update events a feature path produces and where its behavior differs from OFF. They cannot tell whether an admitted ghost belongs to the correct parent. Marks need not represent distinct physical reflections, and a reduction in association count need not represent fewer false positives. Ground-truth labeling would convert that accounting exercise into a perception-accuracy experiment.

K A Focused Publication Boundary

The conference manuscript emphasizes coordinate-consistent virtual-aperture observations, correct uncertainty propagation, a single-use iterated update and a controlled comparison. This extended version preserves the supporting derivations, failures and protocol details. It does not contain a repository backup, raw patches, private production headers or patent claims. Those materials serve different purposes and should not be

used to inflate a scientific paper's length. The remaining empirical questions are precise: compare against a direct tracker with angle observations; classify mixed versus symmetric reflection paths; hold reflector errors fixed across a trajectory; test independently labeled parent/ghost correspondences; and measure estimator consistency as well as error. These are the experiments that can support a stronger future version, not additional adjectives attached to the current result.

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